



COAL PRICE FORECASTING AND PRICE OPTIMIZATION MODEL

NEWCASTLE COAL FUTURES FORECAST JANUARY 2026 - JUNE 2026

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**PROGRAMMING LANGUAGE: PYTHON
COAL BENCHMARK: NEWCASTLE COAL FUTURES**

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Declaration and Acknowledgement

I confirm that I have prepared this assignment as part of the Wipro Data Scientist hiring process. The datasets were collected, cleaned, analysed, and modelled for the purpose of this submission, and the final interpretation and recommendations are based on my own review of the results. I used AI tools only for guidance in structuring the report, improving clarity, and reviewing presentation quality, while I validated the modelling decisions, analysis, assumptions, and business conclusions. This report is intended only for the hiring assessment and should be treated as confidential.

I. Executive Overview

I.1 Project Objective

The objective of this project is to build a coal price forecasting model for the period from January 2026 to June 2026, using historical coal price data and relevant external market indicators through December 2025. The assignment also focuses on translating the forecast into practical price-optimization recommendations for procurement planning, contract timing, inventory management, and price risk monitoring.

1.2 Coal Benchmark Selected

Newcastle Coal Futures was selected as the target benchmark for this study because it is one of the most relevant references for seaborne thermal coal pricing and is closely linked to coal used in power generation and industrial energy demand. Since the assignment required a clearly justified coal benchmark, Newcastle was suitable because it provides a consistent daily price history, has strong market relevance, and reacts to many of the same global drivers used in this model, such as crude oil, natural gas, LNG, freight rates, currency movement, carbon prices, and broader commodity demand. This made it a better fit for a multi-variable forecasting model than a less liquid or less consistently available regional coal price series.

1.3 Data Coverage and Forecast Design

The final historical modelling period covers January 2015 to December 2025. This period provides sufficient long-term market history to capture coal price cycles, volatility, seasonality, and relationships with external market drivers. The forecast horizon is January 2026 to June 2026, with daily coal price forecasts generated first and then summarized into monthly average values for business interpretation and price optimization recommendations.

1.4 Models Compared

Two forecasting approaches were compared in this project. XGBoost was used as the primary machine learning model because it can capture non-linear relationships between coal prices and external market variables. SARIMAX was used as the statistical time-series benchmark to compare the machine learning results against a traditional forecasting approach. This comparison helped evaluate whether a feature-driven machine learning model performed better than a classical time-series model for the coal price forecasting problem.

1.5 Final Model Selected

XGBoost was selected as the final forecasting model because it performed better than SARIMAX during model evaluation and was able to capture the relationship between coal prices and external market variables more effectively. The detailed model performance comparison, including MAE, RMSE, MAPE, and R^2 , is discussed in the model evaluation section of this report.

1.6 Key Forecast Insights

The forecast shows an upward movement in Newcastle coal prices during the first half of 2026. The monthly average forecast increases from approximately 109.20 in January 2026 to around 145.02 in June 2026. This suggests that coal prices may remain under upward pressure during the forecast period, especially if energy demand, freight costs, currency movement, or related commodity prices continue to support higher coal pricing.

1.7 Price Optimization Recommendations

Based on the forecasted upward trend, the recommended strategy is to avoid delaying major coal procurement decisions. Buyers should consider phased procurement during the lower forecasted price months, early contract locking where commercially feasible, and continuous monitoring of key market indicators such as Brent oil, natural gas, JKM LNG, freight rates, and the US Dollar Index. This approach can help reduce exposure to price volatility and support better cost planning for the January to June 2026 period.

2. Business Context and Project Scope

2.1 Background of Coal Price Forecasting

Coal pricing is closely linked with the wider energy and commodity market. It is affected by power demand, industrial activity, fuel substitution, freight movement, currency changes, carbon cost pressure, and global supply conditions. Because of this, coal price forecasting cannot be treated as a simple single-variable time-series problem. A practical forecasting model should consider both the historical movement of coal prices and the external market indicators that influence price behaviour over time.

2.2 Business Need for the Forecast

For companies that purchase coal or operate with coal-linked contracts, price changes can directly affect procurement cost, budgeting, working capital planning, and margin control. Even a small change in coal prices can become material when procurement volumes are large. A forecast can therefore support better decisions on when to buy, whether to lock prices early, how to plan inventory, and when to monitor the market more closely. The main business value of this project is to convert market data into a decision-support view, not just to produce a statistical forecast.

2.3 Project Scope

The scope of this project covers the complete forecasting workflow, starting from data collection and ending with price optimization recommendations. Newcastle Coal Futures was used as the target variable, and external market variables were added to capture broader price drivers. These included Brent oil, natural gas, JKM LNG, US Dollar Index, AUD/USD, Baltic Dry Index, carbon emissions price, and iron ore price. The historical modelling period covers January 2015 to December 2025, and the forecast period covers January 2026 to June 2026.

2.4 Modelling Scope

Two modelling approaches were used and compared. XGBoost was developed as the main machine learning model because it can handle non-linear relationships between coal prices and external variables. SARIMAX was developed as a statistical time-series benchmark to compare the machine learning model against a traditional forecasting approach. The final model selection was based on test-period performance and business suitability for forecasting coal prices over the January to June 2026 period.

2.5 Business Use Cases Covered

The forecast output was designed to support practical business decisions. Daily forecasts provide a detailed view of expected price movement, while monthly average forecasts give a clearer planning view for procurement and management-level discussion. The results can be used for procurement timing, phased buying, contract negotiation, inventory planning, and hedging-related discussions. This makes the project useful not only from a modelling perspective, but also from a commodity cost management perspective.

2.6 Final Deliverables

The final deliverables include the Python notebook, final forecast output dataset, model comparison results, and this report document. The notebook contains the technical implementation, while the report explains the business context, data sources, variable relevance, data quality checks, model selection, forecast interpretation, and price optimization recommendations in a clear and structured manner.

3. Data Collection and Source Strategy

3.1 Data Collection Approach

The historical market data used in this project was collected from Investing.com and downloaded in CSV format. This source was selected because it provides consistent historical price data across commodities, currencies, indices, and futures contracts in a similar structure. For this assignment, that was important because the model required multiple market variables to be cleaned, aligned by date, and combined into one model-ready dataset.

Investing.com was a practical fit for this project because it provided daily historical data for the selected coal benchmark and the external market drivers used in the model. The downloaded files included standard market fields such as Date, Price, Open, High, Low, Volume, and Change percentage. This made the source suitable for building a reproducible forecasting workflow without relying on paid data feeds or commercial APIs.

3.2 Target Variable Source

Newcastle Coal Futures was used as the target variable and was collected from Investing.com historical data. It was selected because it is a relevant benchmark for seaborne thermal coal pricing and provides daily price movement suitable for forecasting. The closing price of Newcastle Coal Futures was used as the main target column for model development.

3.3 Independent Variables Collected

Eight external variables were collected from Investing.com and used along with the target coal price series. These variables were selected because they represent the major market forces that can influence coal pricing, including energy substitution, global fuel demand, currency movement, freight market conditions, carbon cost pressure, and broader industrial commodity demand.

The independent variables used in the final dataset were Brent oil, natural gas, US Dollar Index, AUD/USD, Baltic Dry Index, carbon emissions futures, JKM LNG futures, and iron ore futures.

3.4 Free vs Paid Data Source Classification

All datasets used in this project were collected from Investing.com historical data pages and downloaded manually as CSV files. No paid API, subscription-based data plugin, or commercial data feed was used in the modelling workflow. This made the data sourcing approach cost-effective and easy to reproduce for an assignment-level forecasting project.

The main limitation of this approach is that manual CSV downloads are less automated than a paid API or institutional data platform. Paid sources such as Bloomberg, Refinitiv, S&P Global, or exchange-level commodity data providers may offer stronger licensing support, automated pipelines, and richer metadata. However, for the purpose of this assignment, Investing.com was sufficient because it provided the required daily historical market data across the target variable and selected independent variables.

3.5 Data Frequency and Coverage

The datasets were collected at daily frequency because the assignment required daily granularity. Daily data also provided enough observations for model training and allowed the final forecast to be reviewed at both daily and monthly levels. Since different markets have different holidays and trading calendars, all datasets were aligned using the Date column after loading. Newcastle Coal Futures was used as the base series during merging because it was the target variable.

3.6 Data Source Summary Table

Table 3.1: Data Sources, Variable Purpose, and Access Classification

Variable	Dataset Used	Role in Model	Source / Access Method	Free or paid	Frequency
Newcastle Coal Futures	Newcastle Coal Futures Historical Data	Target variable	Investing.com Link [Dataset]	Free public access	Daily
Brent Oil Futures	Brent Oil Futures Historical Data	Independent variable	Investing.com Link [Dataset]	Free public access	Daily
Natural Gas Futures	Natural Gas Futures Historical Data	Independent variable	Investing.com Link [Dataset]	Free public access	Daily
US Dollar Index	US Dollar Index Historical Data	Independent variable	Investing.com Link [Dataset]	Free public access	Daily
AUD/USD	AUD_USD Historical Data	Independent variable	Investing.com Link [Dataset]	Free public access	Daily
Baltic Dry Index	Baltic Dry Index Historical Data	Independent variable	Investing.com Link [Dataset]	Free public access	Daily
Carbon Emissions Futures	Carbon Emissions Futures Historical Data	Independent variable	Investing.com Link [Dataset]	Free public access	Daily
JKM LNG Futures	LNG Japan/Korea Marker PLATTS Future Historical Data	Independent variable	Investing.com Link [Dataset]	Free public access	Daily
Iron Ore Futures	Iron Ore Fines 62% Fe CFR Futures Historical Data	Independent variable	Investing.com Link [Dataset]	Free public access	Daily

3.7 Data Source Reliability and Limitations

The selected data source provided historical market data for all variables required in the project. However, some limitations were observed during data preparation. Missing observations can occur because of holidays, non-trading days, or differences in trading calendars across commodities, currencies, and indices. In addition, volume data contained a large number of missing values and was therefore less useful for modelling purposes. As a result, the forecasting workflow primarily relied on price-related fields such as Price, Open, High, Low, and Change percentage. These issues were addressed during the data cleaning and preprocessing stage before model development.

4. Variables Used and Their Relevance

4.1 Variable Selection Approach

The variables used in this project were selected based on their practical relationship with coal price movement. Coal is part of the wider energy and commodity market, so its price is affected not only by its own historical movement, but also by changes in fuel prices, shipping costs, currency movement, environmental pricing, and industrial demand.

For this reason, the final dataset was built using Newcastle Coal Futures as the target variable and a selected set of external market indicators as independent variables. The selection was kept focused on variables that had daily data availability and a clear market explanation. This was important because the model was designed to generate daily forecasts, and using lower-frequency or weakly related variables could have reduced the consistency of the modelling dataset.

4.2 Target Variable: Newcastle Coal Futures

Newcastle Coal Futures was used as the target variable for the forecasting model. This benchmark was selected because it represents thermal coal pricing in the seaborne market and is closely linked with coal used for power generation and industrial energy demand.

The closing price of Newcastle Coal Futures was used as the main value to be predicted. This gave the model a consistent daily price series from January 2015 to December 2025 and made it suitable for both time-series modelling and machine learning-based forecasting. Since the assignment required the candidate to justify the coal benchmark, Newcastle was an appropriate choice because it is market-relevant, available at daily frequency, and directly aligned with the objective of forecasting coal prices.

4.3 Brent Oil Futures

Brent oil was included because it reflects the broader global energy market. Coal and crude oil are not always direct substitutes, but both are influenced by energy demand, geopolitical risk, inflationary pressure, and commodity market cycles. When oil prices rise, it often indicates tighter energy market conditions, which can also affect the pricing environment for other fuels.

In this model, Brent oil was used to capture whether coal prices were moving independently or as part of a wider energy market trend. This made it a useful external driver for understanding energy-price pressure around coal.

4.4 Natural Gas Futures

Natural gas was included because it has a stronger substitution relationship with coal, especially in power generation. In several markets, electricity producers compare coal and gas costs when deciding fuel usage. When natural gas prices increase, coal can become relatively more attractive as a fuel source, which may support coal demand and coal prices. When natural gas prices fall, coal can face demand pressure.

This variable was therefore important for capturing fuel-switching behaviour and the relationship between coal and gas in the energy market.

4.5 JKM LNG Futures

JKM LNG was included because it represents LNG pricing in the Asian energy market. This is relevant because Newcastle coal is closely connected with the Asia-Pacific thermal coal trade. LNG and coal can compete as energy sources, particularly for power generation and industrial fuel demand.

Including JKM LNG helped the model capture regional energy market pressure that may not be fully reflected by natural gas futures alone. A rise in LNG prices can improve the relative attractiveness of coal for some buyers, while lower LNG prices may reduce that support.

4.6 US Dollar Index

The US Dollar Index was included because coal and other globally traded commodities are strongly influenced by currency movement. A stronger US dollar can make dollar-priced commodities more expensive for buyers using other currencies, which can affect demand and pricing. A weaker dollar can improve affordability for international buyers and may support commodity prices.

For this project, the US Dollar Index was used to capture the broader currency environment affecting global commodity trade.

4.7 AUD/USD Exchange Rate

AUD/USD was included because Australia is one of the major coal-exporting countries, and Newcastle coal is linked with the Australian export market. Movement in the Australian dollar can influence export competitiveness, producer revenue, and pricing conditions for Australian coal.

This variable was relevant because it connects the forecasting model to the currency exposure of the coal export region. A weaker Australian dollar can support export competitiveness, while a stronger Australian dollar can affect margins and pricing pressure.

4.8 Baltic Dry Index

The Baltic Dry Index was included as a freight market indicator. Coal is a high-volume commodity and a large part of global coal trade moves through seaborne routes. Freight rates therefore affect the delivered cost of coal, especially for importers.

This variable helped the model capture shipping and transport cost pressure. Even if the coal benchmark price does not move sharply, changes in freight conditions can influence buyer behaviour and the overall landed cost of coal.

4.9 Carbon Emissions Futures

Carbon emissions futures were included because carbon pricing affects the economics of coal usage. Coal is carbon-intensive, so higher carbon prices can increase the cost of coal-based energy generation in markets where carbon regulation applies. Carbon prices also reflect policy pressure and transition risk around fossil fuel use.

This variable was included to capture environmental cost pressure and the policy-linked market signal that can influence coal demand over time.

4.10 Iron Ore Futures

Iron ore was included as a broader industrial commodity indicator. While iron ore does not directly determine thermal coal prices, it is linked with industrial activity, infrastructure demand, and steel-sector conditions. These factors are also connected with wider commodity demand, especially in Asia.

In this project, iron ore was used as a proxy for industrial commodity momentum. It helped provide additional market context when coal prices were influenced by broader demand conditions rather than only energy-specific factors.

4.11 Overall Variable Relevance Summary

The final set of variables was selected to give the model a balanced view of the main market forces that can influence Newcastle coal prices. Brent oil, natural gas, and JKM LNG captured energy market movement and fuel substitution pressure. The US Dollar Index and AUD/USD captured currency-related effects on commodity pricing and Australian coal export conditions. The Baltic Dry Index captured freight market pressure, carbon emissions futures captured environmental cost pressure, and iron ore captured wider industrial commodity demand.

These variables were not selected randomly. Each one represents a practical market signal that can help explain coal price movement. The relationship between each variable and coal prices may change across different market periods, but together they provided a stronger modelling base than relying only on historical coal prices.

5. Dataset Accuracy, Quality, and Validation

5.1 Raw Data Inspection

The raw data used in this project consisted of nine historical market datasets: Newcastle Coal Futures, Brent Oil Futures, Natural Gas Futures, US Dollar Index, Baltic Dry Index, AUD/USD, Carbon Emissions Futures, JKM LNG Futures, and Iron Ore Futures. Each file was loaded from CSV format and reviewed before cleaning.

During the initial inspection stage, the datasets were checked for row count, column structure, sample records, data types, missing values, date coverage, and duplicate dates. This confirmed that all required files were loaded correctly and that the raw structure was suitable for preparing a combined forecasting dataset.

Dataset	Raw Records
Newcastle Coal Futures	2,955
Brent Oil Futures	2,960
Natural Gas Futures	3,016
US Dollar Index	2,983
Baltic Dry Index	2,853
AUD/USD	2,993
Carbon Emissions Futures	2,946
JKM LNG Futures	2,877
Iron Ore Futures	2,883

5.2 Date Coverage and Forecast Period Validation

Date coverage was checked separately for every dataset before merging. This was important because the assignment required historical data up to December 2025 and a forecast period from January 2026 to June 2026. The Newcastle Coal Futures dataset, which was used as the target series, covered 2 January 2015 to 19 June 2026 in the downloaded file. For modelling, the historical period was kept up to 31 December 2025. The available forecast comparison period contained 120 trading-day records from 2 January 2026 to 19 June 2026.

5.3 Missing Values, Duplicates, and Data Cleaning

Missing values were reviewed during raw data inspection, after merging, and again after feature engineering. The Date and Price columns were complete in the raw datasets, which was important because Date was used for merging and Price was the main modelling field.

The main missing-value issue was found in the Volume column. Since Volume was not consistently available across all datasets and was not required for the final modelling approach, it was excluded from the modelling workflow. After merging the datasets, some missing values appeared because different markets

follow different holidays and trading calendars. These gaps were handled using forward fill, which carried the latest available market value forward.

Duplicate Date records were also checked before merging and after master dataset creation. The final cleaned master dataset had zero missing values and zero duplicate dates.

5.4 Data Format and Date Alignment

The raw files contained market values in display-friendly format, so they were converted before modelling. The Date column was converted into datetime format, and market fields such as Price, Open, High, Low, and Change percentage were converted into numeric format after removing commas and percentage symbols.

Column names were standardized using clear prefixes for each dataset, such as *newcastle_coal_price* and *brent_oil_price*. This made the merged dataset easier to read and avoided confusion because the raw files had similar column names.

Newcastle Coal Futures was used as the base dataset during merging because it was the target variable. All other market variables were aligned to this base using the Date column. The merged master dataset contained 2,955 rows and 46 columns before final feature selection.

5.5 Feature-Level Validation and Volatility Review

Before modelling, the merged dataset was reviewed again to decide which fields should be retained as model inputs. Open, High, Low, and Change percentage columns were checked against the corresponding closing price columns. Since Open, High, and Low were highly correlated with closing prices, the final modelling dataset focused mainly on closing price variables, calendar features, and coal lag features.

Calendar features such as year, month, and quarter were added. Coal lag features were also created to capture short-term and monthly historical movement. The missing values created at the beginning of the dataset due to lag features were filled using backward fill.

No formal outlier removal, capping, or winsorization was applied. This was intentional because sharp movements in coal and energy prices can represent genuine market behaviour. Instead, volatility and abnormal price movements were reviewed during EDA using trend, monthly movement, yearly movement, and distribution checks.

5.6 Final Dataset Quality Summary and Limitations

The final feature-ready dataset contained 2,955 rows and 16 columns. It had zero missing values and zero duplicate dates. The historical model data contained 2,835 records up to December 2025, while the available forecast-period data contained 120 records from January to June 2026.

The final dataset was suitable for chronological splitting, model development, model evaluation, and coal price forecasting. The main assumptions were that forward fill was appropriate for non-trading-day gaps, Volume could be excluded due to inconsistent availability, and genuine market volatility should be retained rather than removed.

The main limitation is that the data was collected from publicly available CSV downloads rather than an automated institutional data feed. This made the workflow cost-effective and suitable for the assignment, but future production use would benefit from an automated API or licensed market data provider for stronger governance and regular updates.

6. Exploratory Data Analysis (EDA)

6.1 EDA Preparation and Objective

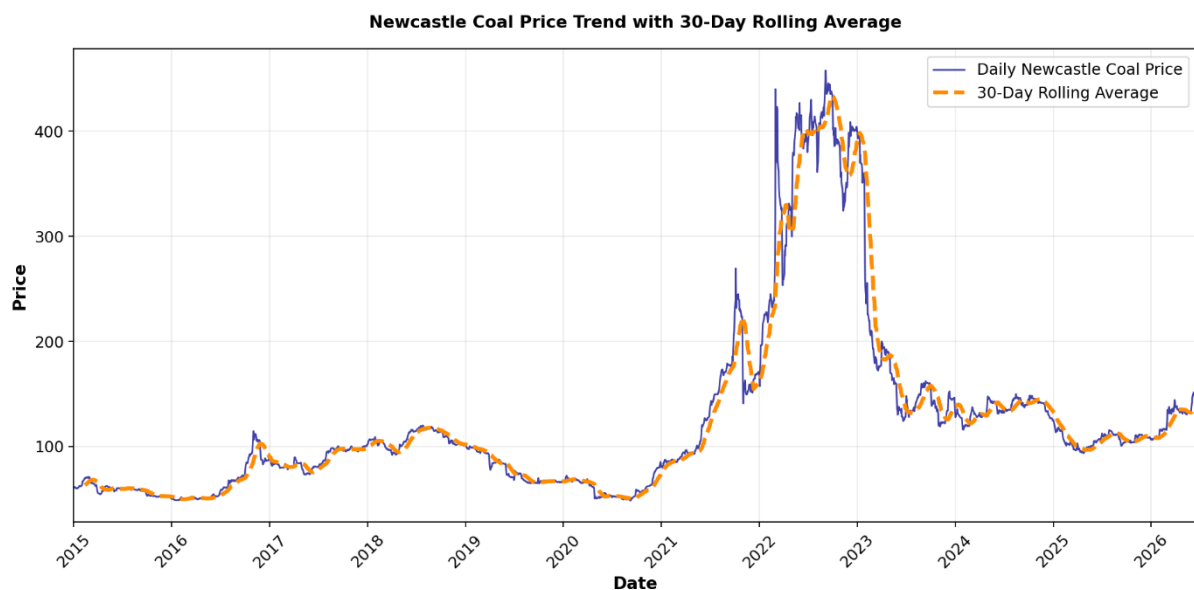
The exploratory data analysis was carried out after the cleaned master dataset was prepared. A separate EDA copy was created from the master dataset so that the original cleaned data remained unchanged for the feature engineering and modelling stages.

Before analysing the data, the Date column was converted into the correct datetime format and the records were sorted in chronological order. This was necessary because the project is a time-series forecasting problem, where the sequence of observations is important.

The purpose of this EDA was to understand how Newcastle coal prices moved over the historical period and how the selected market variables behaved alongside coal prices. The analysis focused on price trends, external market movements, correlation with independent variables, monthly price behaviour, yearly price movement, and volatility. These checks helped build a clear view of the data before moving into feature engineering and model development.

6.2 Newcastle Coal Price Trend

The Newcastle coal price trend was analysed using the daily closing price and a 30-day rolling average. The daily line shows actual market movement, while the rolling average helps smooth short-term fluctuations and highlights the broader direction.

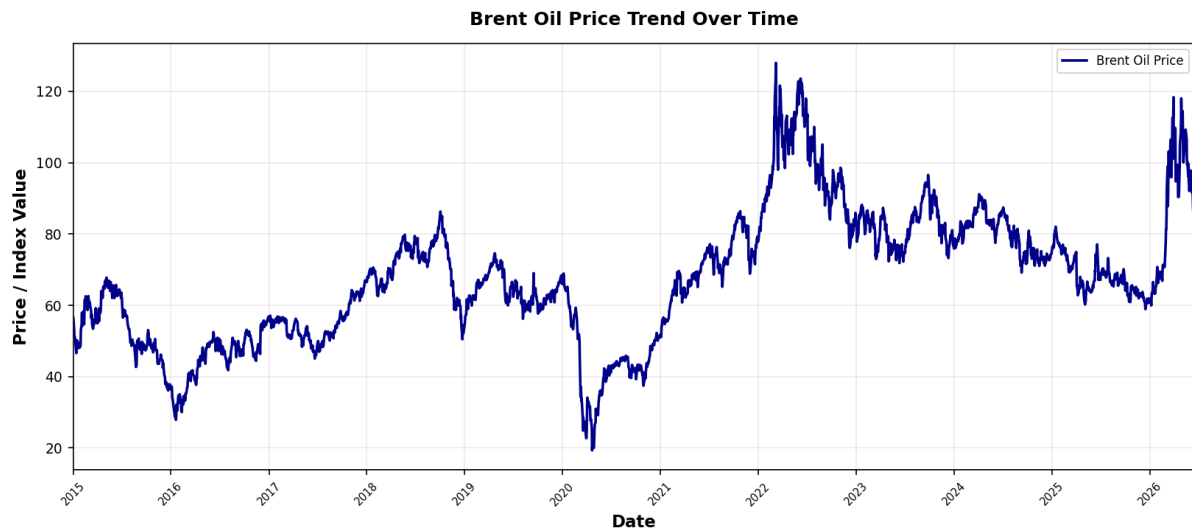


From 2015 to 2020, coal prices mostly stayed in a lower and relatively stable range. A clear shift appeared from 2021, followed by a sharp price surge in 2022, where prices moved above 400 at the peak. After this high-volatility period, prices corrected during 2023 and remained more stable through 2024 and most of 2025, although still above pre-2021 levels.

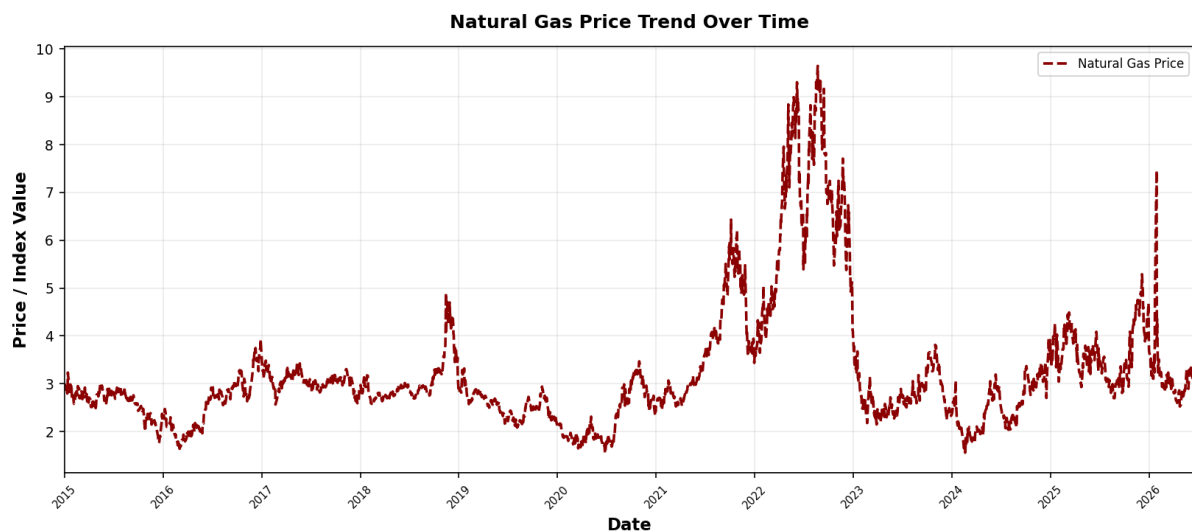
This trend review showed that Newcastle coal prices had clear regime changes across the historical period. For modelling, this was important because the model needed to learn both stable price periods and sharp market movements. These movements were retained in the dataset because they represent real commodity market behaviour rather than simple data errors.

6.3 External Market Variable Trends

The external market variables were reviewed separately because each variable represents a different market factor and uses a different scale. Brent oil, natural gas, JKM LNG, carbon emissions, and iron ore are price-based commodity variables, while the US Dollar Index is a currency index, AUD/USD is an exchange rate, and the Baltic Dry Index is a freight market indicator. Reviewing them separately made the trend analysis clearer and avoided misleading comparison on a single chart.

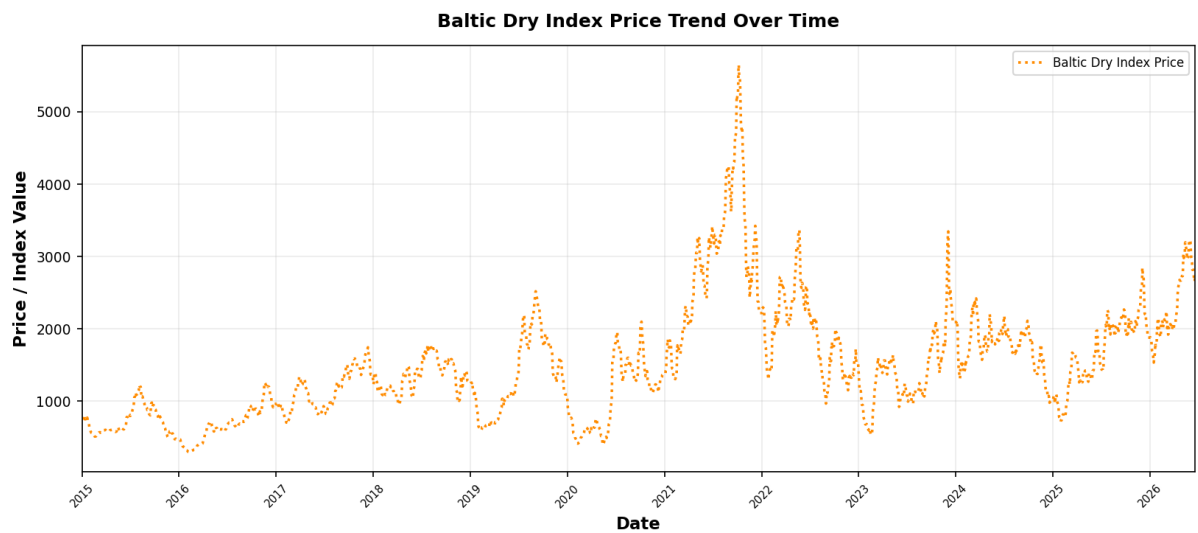
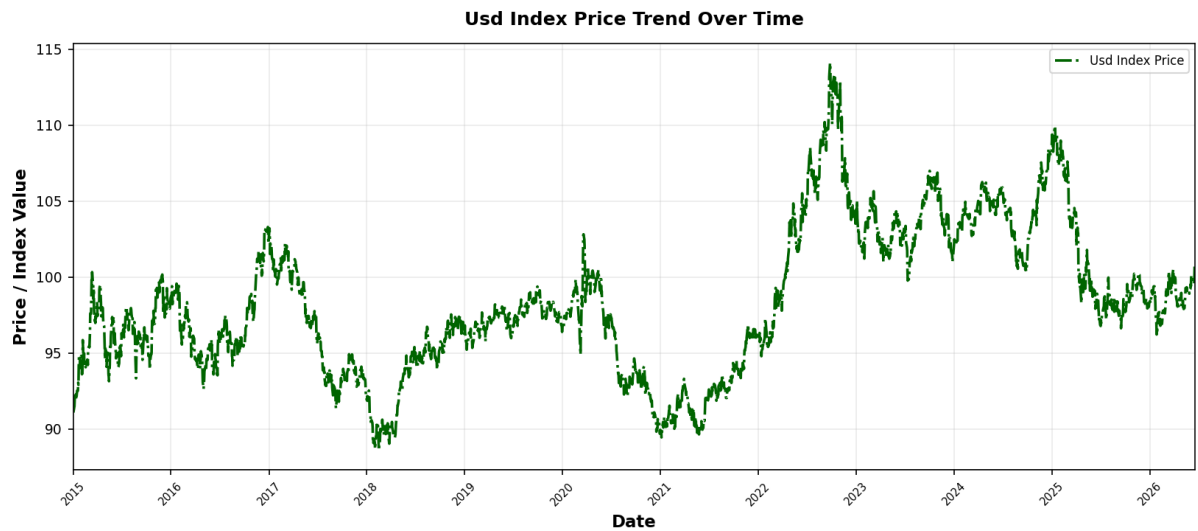


Brent oil showed clear energy market cycles across the historical period, with a sharp fall around 2020, a strong recovery during 2021–2022, and another upward movement toward 2026. This supported its use as a broader energy market indicator for coal price forecasting.

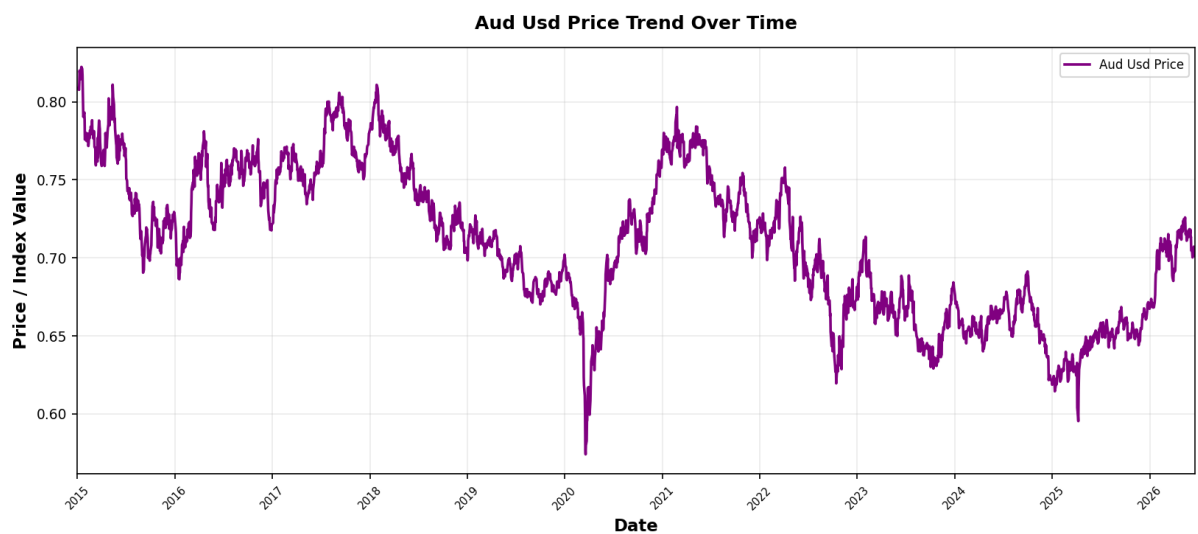


Natural gas showed strong volatility, especially during 2021–2022, when prices moved sharply higher before correcting in 2023. Since natural gas can act as a substitute for coal in power generation, this trend was important for understanding fuel-switching pressure.

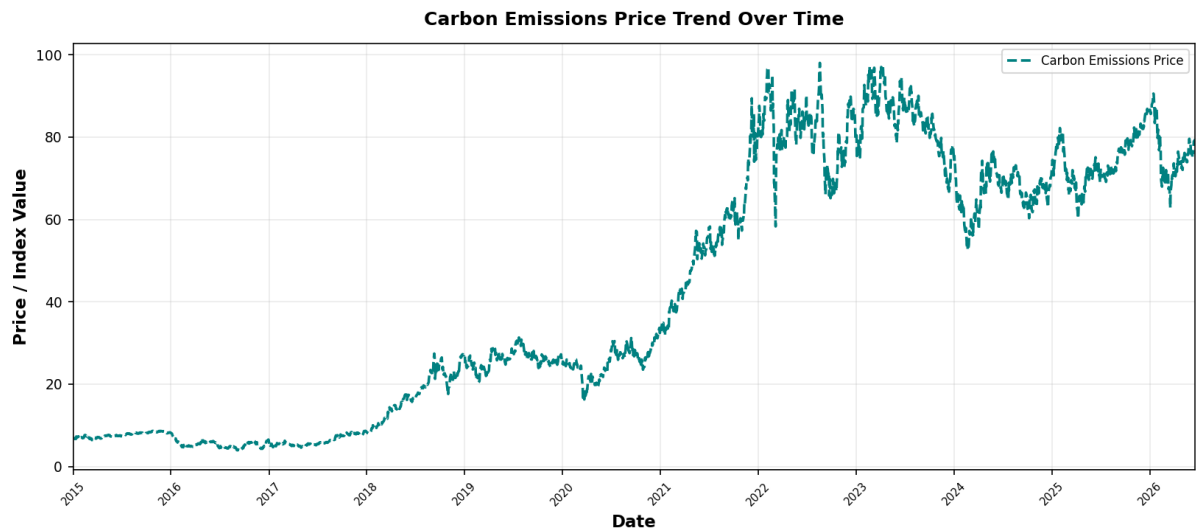
The US Dollar Index showed several periods of strength, especially around 2022 and again around 2024–2025. Since globally traded commodities are influenced by dollar movement, this variable helped capture the currency environment around coal and other commodity prices.



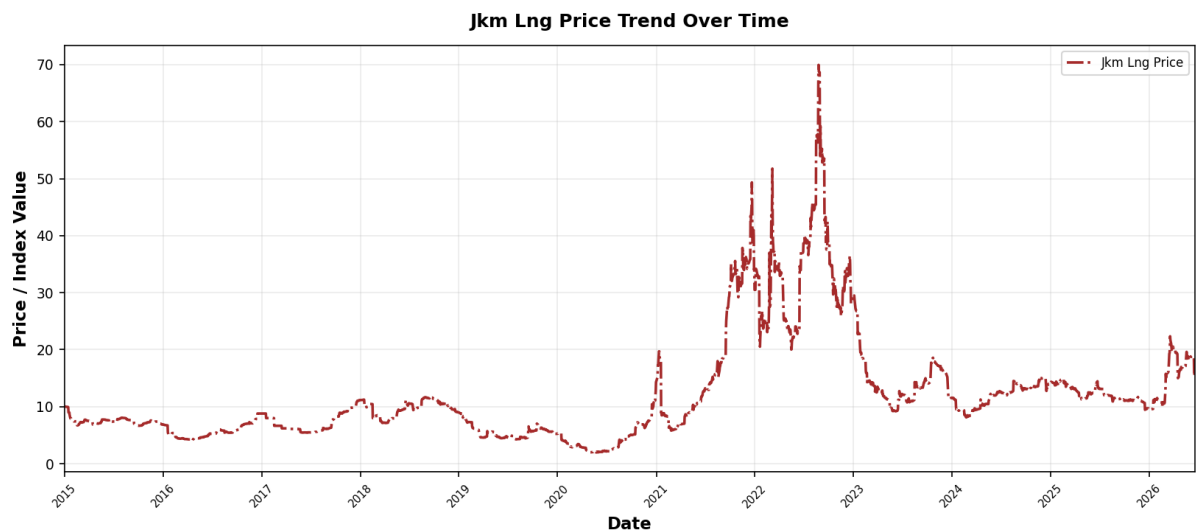
The Baltic Dry Index showed large swings across the period, with a major spike around 2021 and further movement toward 2026. As coal is a high-volume seaborne commodity, freight market pressure can affect delivered coal costs and buyer behaviour.



AUD/USD showed a broad weakening trend after 2018, with some recovery toward 2026. This was relevant because Newcastle coal is linked to the Australian export market, and currency movement can influence export competitiveness and pricing conditions.

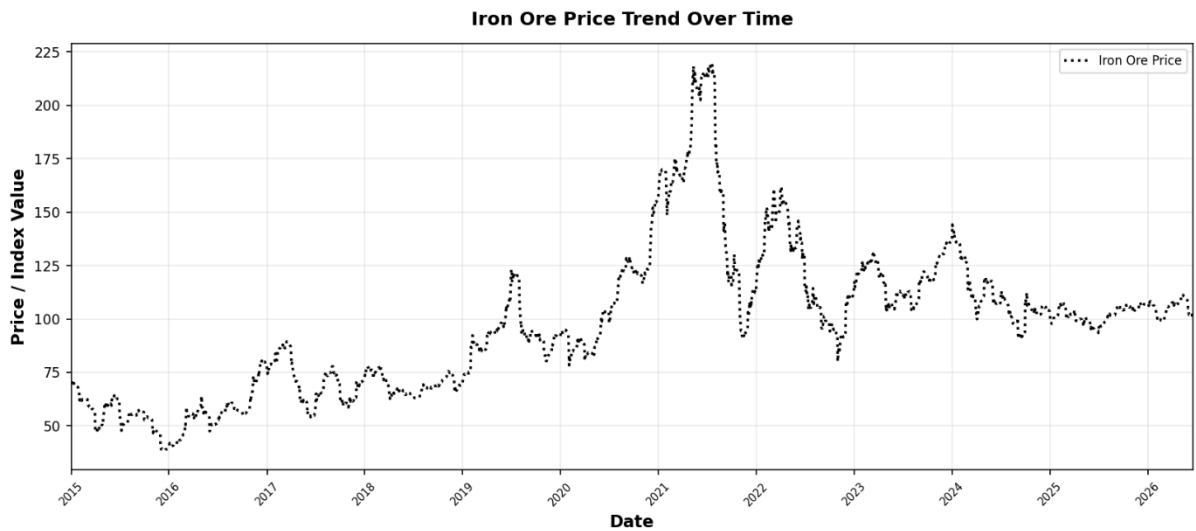


Carbon emissions prices increased strongly after 2020 and stayed at a higher level compared with the earlier years. This trend was important because carbon pricing reflects environmental cost pressure and policy risk around coal usage.



JKM LNG showed a sharp price surge during 2021–2022, followed by a correction and later recovery. This variable was relevant because LNG is an important energy market indicator in Asia, where Newcastle coal also has strong market relevance.

Iron ore showed a strong upward cycle around 2020–2021, followed by a correction and more stable movement afterward. It was used as a broader industrial commodity indicator because it reflects steel, infrastructure, and industrial demand conditions.

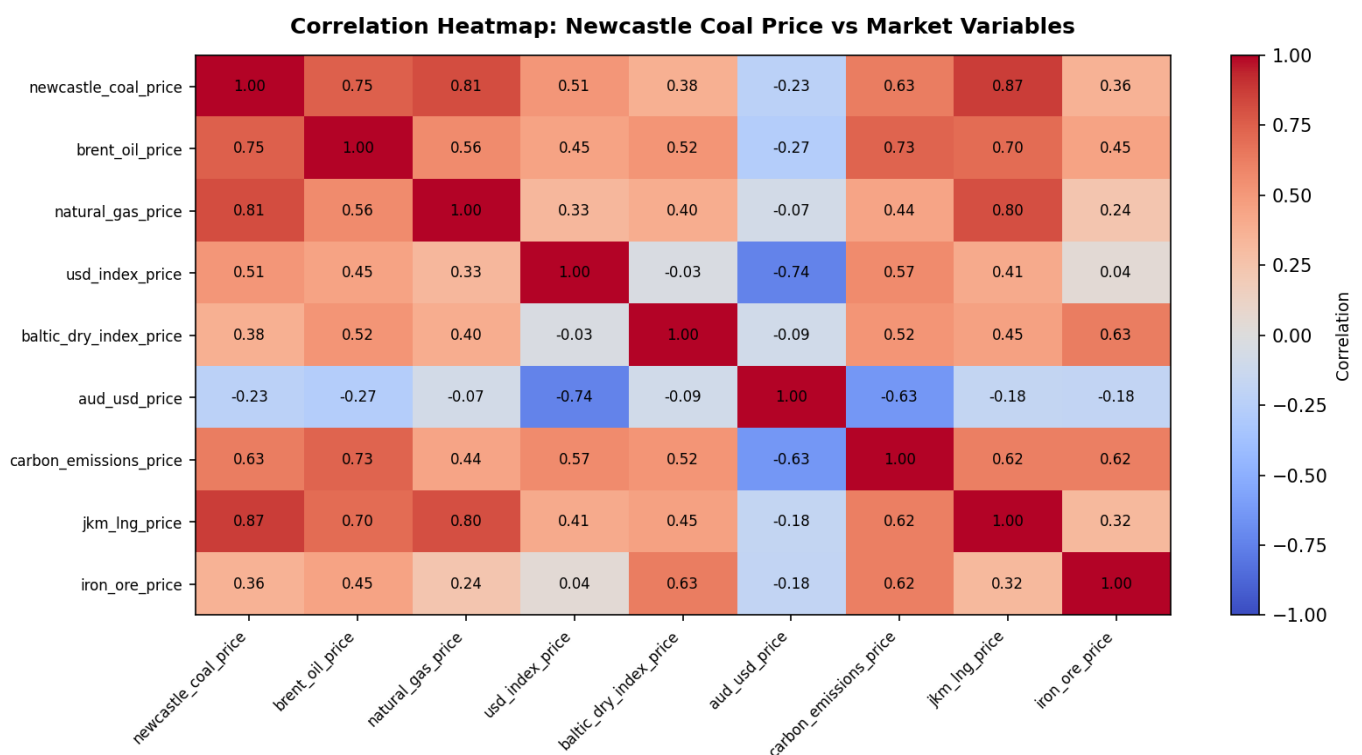


Overall, the trend review showed that the selected variables captured different parts of the coal pricing environment: energy market cycles, fuel substitution, currency movement, freight cost pressure, carbon pricing, and industrial demand. This supported their use as external inputs for the forecasting model.

6.4 Correlation Analysis

A correlation heatmap was created to understand the strength and direction of the relationship between Newcastle coal prices and the selected market variables. This step was useful before modelling because it showed which external variables moved more closely with coal prices over the historical period.

The strongest positive correlation with Newcastle coal price was observed for JKM LNG, followed by natural gas and Brent oil. This makes business sense because all three variables are linked to the broader energy market. JKM LNG and natural gas are especially relevant because they can act as alternative fuel indicators for coal, particularly in power generation and Asian energy markets. Brent oil also showed a strong positive relationship, suggesting that coal prices were often moving in line with wider energy market cycles.



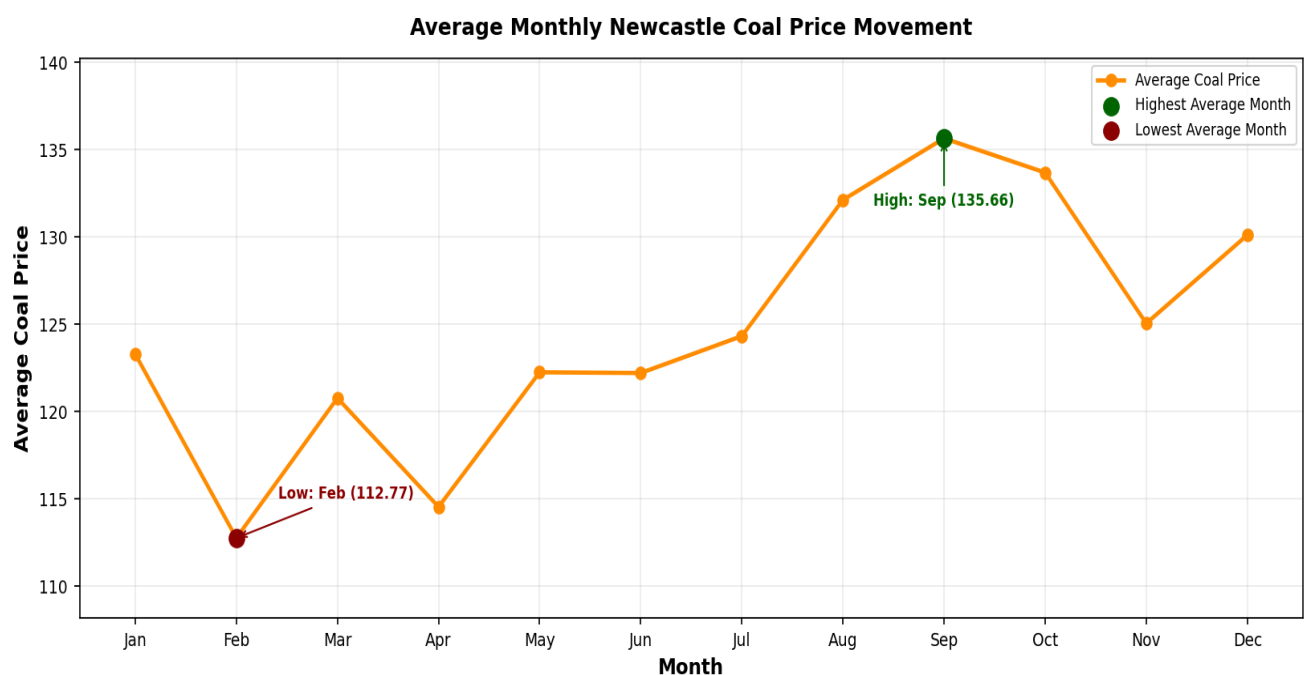
Carbon emissions also showed a meaningful positive correlation with coal prices. This does not mean carbon prices directly increase coal prices, but it indicates that both variables moved through a similar high-price period, especially after 2020 when energy and environmental markets became more volatile. The US Dollar Index showed a moderate positive relationship, while the Baltic Dry Index and iron ore had weaker but still positive relationships with coal prices.

AUD/USD showed a negative correlation with Newcastle coal prices. This is reasonable because currency movements can affect Australian export pricing conditions, and a weaker Australian dollar may support export competitiveness. However, the relationship should be interpreted carefully because currency effects can change depending on market conditions.

The heatmap also showed correlation between some independent variables themselves, such as natural gas with JKM LNG, Brent oil with carbon emissions, and the US Dollar Index with AUD/USD. This confirmed that the selected variables were connected through broader market cycles. For modelling, this was acceptable because the final machine learning model was able to use multiple market signals together, but the relationships were interpreted with caution. Correlation was used as an exploratory check, not as proof of direct causation.

6.5 Monthly Coal Price Movement

Monthly coal price movement was reviewed to understand whether Newcastle coal prices showed any visible month-wise pattern across the historical data. The analysis used the average coal price for each calendar month and highlighted the months with the highest and lowest average values.

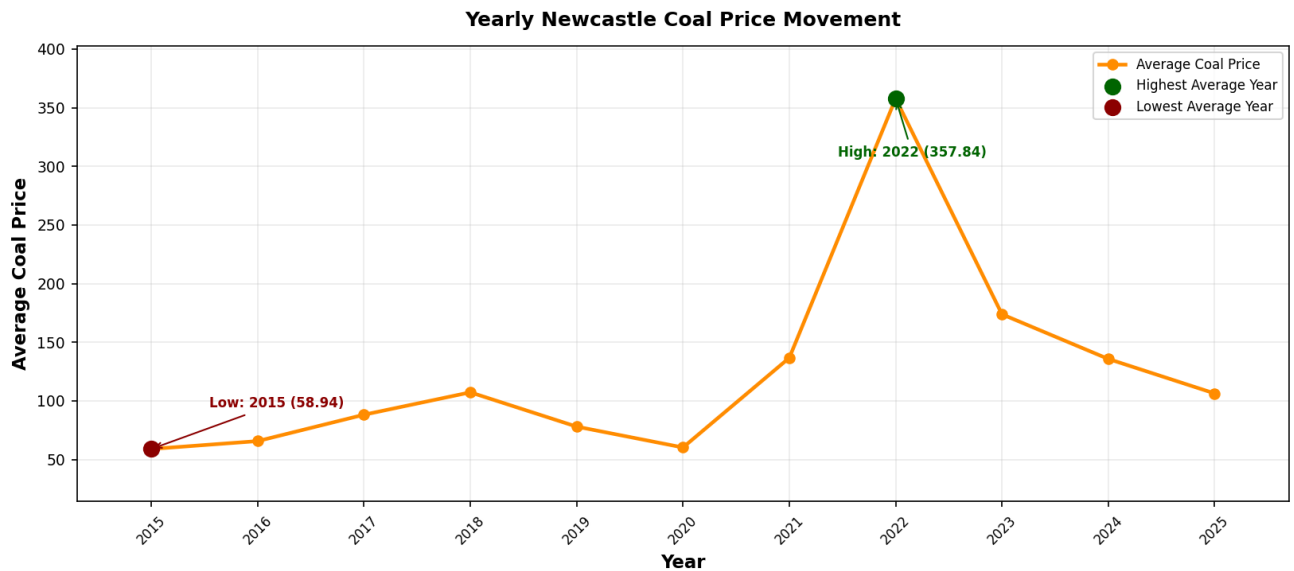


The monthly view showed that September had the highest average Newcastle coal price at 135.66, while February had the lowest average price at 112.77. Prices were generally stronger from August to October, while February and April appeared weaker compared with the rest of the year. This suggests that coal prices may have some seasonal behaviour, although the pattern should be interpreted carefully because coal prices are also affected by market shocks, energy demand, freight movement, and global commodity cycles.

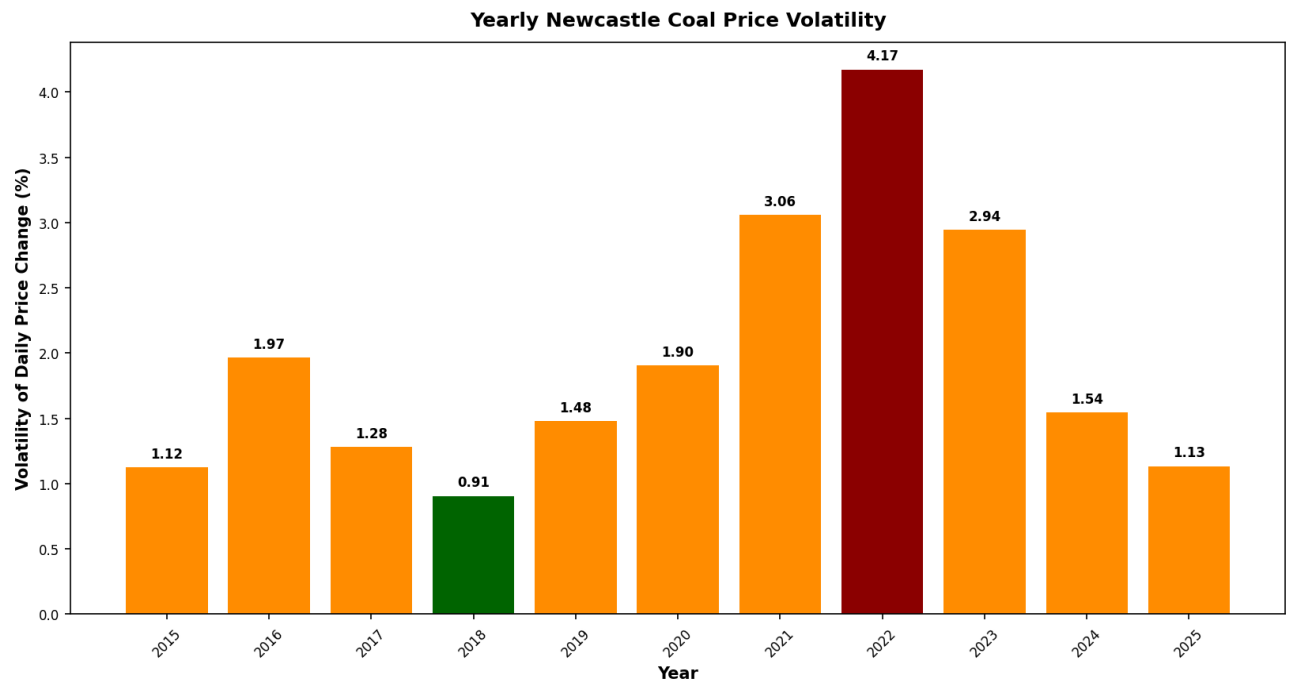
This monthly analysis was useful for the modelling stage because it supported the inclusion of calendar features such as month and quarter. It also helped in business interpretation, as monthly averages are easier to connect with procurement planning and contract timing than daily price movements alone.

6.6 Yearly Price Movement and Volatility

The yearly coal price analysis was prepared using historical data up to December 2025, keeping the January to June 2026 period separate for forecasting. This view helped identify how Newcastle coal prices changed across different market periods.



The yearly average price chart shows that Newcastle coal prices were relatively low during 2015–2020, with 2015 having the lowest average price at 58.94. Prices increased sharply in 2021 and reached the highest yearly average in 2022 at 357.84. After 2022, prices corrected strongly, with the yearly average falling to 173.80 in 2023, 135.72 in 2024, and 106.31 in 2025.



The volatility chart shows a similar pattern. Volatility was highest in 2022 at 4.17%, followed by 2021 and 2023. This confirms that the 2021–2023 period was not only a high-price period, but also a high-risk period with larger daily price movements. By comparison, 2024 and 2025 showed lower volatility, suggesting that the market became more stable after the extreme price movements seen earlier.

This analysis was important for modelling because it showed that the historical data contains both stable years and abnormal high-volatility years. The model therefore needed to learn from different market conditions rather than only from recent stable periods.

6.7 Key EDA Findings

- Newcastle coal prices were relatively stable during 2015–2020, before a clear price breakout from 2021 onwards.
- The strongest price spike occurred in 2022, where the yearly average coal price reached 357.84, making it the highest-price year in the historical period.
- After the 2022 peak, coal prices corrected during 2023 and continued to move into a more stable range during 2024 and 2025.
- Yearly volatility was also highest in 2022 at 4.17%, confirming that this period had both high prices and high daily price movement.
- Monthly analysis showed stronger average coal prices during August to October, with September having the highest monthly average price of 135.66.
- February had the lowest average monthly coal price at 112.77, indicating a weaker average month across the historical period.
- Correlation analysis showed that JKM LNG, natural gas, and Brent oil had the strongest positive relationships with Newcastle coal prices.
- AUD/USD showed a negative relationship with Newcastle coal prices, which supports its use as a relevant currency-related market variable.
- The EDA confirmed that coal prices are influenced by both historical price behaviour and external market drivers.
- These findings supported the use of calendar features, coal lag features, and external market variables in the modelling stage.

7. Data Processing and Feature Engineering

7.1 Model-Ready Dataset Preparation

After completing data cleaning and EDA, the cleaned master dataset was converted into a model-ready dataset. A separate modelling dataframe was created from the cleaned master dataset and sorted by Date to maintain the correct time order. A quick quality check was performed before feature creation, confirming that the dataset had no missing values and no duplicate dates.

This step was important because forecasting models should be trained on properly ordered time-series data. The earlier raw cleaning, date conversion, numeric conversion, and merging steps were not repeated here because they had already been completed during master dataset preparation.

7.2 Feature Selection Logic

The merged dataset contained Price, Open, High, Low, and Change percentage columns for each market variable. Before selecting the final model inputs, Open, High, Low, and Change percentage columns were compared with their respective closing price columns.

The review showed that Open, High, and Low were highly correlated with closing prices across the market variables. To avoid adding repeated information to the model, the final modelling dataset retained the Date column and the closing price columns only. This kept the feature set simpler, cleaner, and easier to interpret.

The closing price variables retained were Newcastle coal, Brent oil, natural gas, US Dollar Index, Baltic Dry Index, AUD/USD, carbon emissions, JKM LNG, and iron ore.

7.3 Currency and Calendar Features

AUD/USD was converted from decimal exchange-rate format into cents for easier interpretation. For example, a value such as 0.8087 was converted to 80.87. This was only a scale conversion and did not change the meaning of the variable.

Calendar features were then created from the Date column. The final dataset included year, month, and quarter. These features were added because coal prices showed visible monthly and yearly patterns during EDA, and the model needed simple calendar signals to capture time-related behaviour.

7.4 Coal Lag Features

Coal lag features were created to capture recent historical price movement in the target variable. Three lag features were added: one-day lag, seven-day lag, and thirty-day lag. These features helped the model learn short-term and monthly price behaviour from the Newcastle coal price series.

The lag features naturally created missing values at the start of the dataset because previous coal prices were not available for the first few records. These initial missing values were filled using backward fill. Mean or maximum filling was not used because those methods could create artificial price values that do not respect the time-series structure.

7.5 Historical and Forecast Period Split

The feature-ready dataset was split into historical and forecast periods. The historical model data covered observations up to 31 December 2025. The January to June 2026 period was kept separately for forecast output and comparison.

The available forecast-period records started from 2 January 2026 and ended on 19 June 2026. These 120 records represent available trading days, not full calendar days, because commodity and financial markets do not trade on weekends and exchange holidays.

7.6 Final Feature-Ready Dataset

The final feature-ready dataset contained 2,955 rows and 16 columns. It included the target variable, selected external market price variables, calendar features, and coal lag features. The final dataset had zero missing values and zero duplicate dates.

The historical model dataset contained 2,835 records, while the forecast-period dataset contained 120 records. The completed feature-ready dataset was saved as `model_ready_coal_forecasting_data.csv` and used for model development and evaluation.

8. Model Development Methodology

8.1 Forecasting Strategy

The modelling approach was designed as a supervised time-series forecasting problem. The target variable was `newcastle_coal_price`, and the model inputs included external market price variables, calendar features, and coal lag features. The final input set contained 14 features: Brent oil, natural gas, US Dollar Index, Baltic Dry Index, AUD/USD, carbon emissions, JKM LNG, iron ore, year, month, quarter, coal lag 1, coal lag 7, and coal lag 30.

A chronological split was used because the objective was to forecast future coal prices from past information. The model was trained on older data, validated on the next period, tested on unseen 2025 data, and then used for the January to June 2026 forecast period. This approach avoided random shuffling and kept the workflow close to a real forecasting situation.

8.2 Baseline Model

A simple baseline model was created using the previous available Newcastle coal price as the prediction. This was included as a reference point before evaluating more advanced models.

The baseline was useful because coal prices often show short-term persistence. If a model cannot improve on a simple previous-day price assumption, then the added modelling complexity is difficult to justify. For this reason, the baseline helped set a practical minimum benchmark for model performance.

8.3 Machine Learning Model Approach

XGBoost was used as the main machine learning model. It was selected because it can capture non-linear relationships between coal prices and multiple external market drivers. This was important for this project because coal prices are influenced by a mix of energy prices, currency movement, freight conditions, carbon prices, industrial demand, and recent coal price behaviour.

The XGBoost model used the engineered feature set, including external variables, calendar features, and coal lag features. The model was first trained on the training period and evaluated on the validation period to select the best parameter setting. After selecting the best setting, the model was trained again using training and validation data together and then tested on the unseen 2025 period.

8.4 Time-Series Model Approach

SARIMAX was used as the statistical time-series comparison model. It was selected because it can model historical coal price behaviour while also using external market variables as exogenous inputs. This made it a suitable comparison against the XGBoost machine learning approach.

The SARIMAX model was tested with selected order combinations and evaluated on the validation period. The model used the same external market drivers so that the comparison with XGBoost remained fair. The purpose of including SARIMAX was to check whether a traditional statistical forecasting model could compete with the feature-driven machine learning model.

8.5 Hyperparameter Tuning and Cross-Validation Methodology

For XGBoost, a small parameter grid was tested using different values of `n_estimators`, `max_depth`, and `learning_rate`. The best setting was selected based on validation RMSE because RMSE gives higher penalty to larger forecast errors.

For SARIMAX, a few order combinations were tested and compared using the same validation approach. The best SARIMAX order was also selected based on validation performance.

TimeSeriesSplit was used to check model stability over different time periods. This was more appropriate than random cross-validation because time-series data must preserve chronological order. In each split, earlier observations were used to predict a later period, which follows the same logic as the actual forecasting task.

8.6 Evaluation Metrics Used

The models were evaluated using MAE, RMSE, MAPE, and R² score. MAE measured the average error in coal price units. RMSE was used to give stronger penalty to larger errors. MAPE expressed the error in percentage terms, which is easier to interpret from a business point of view. R² showed how much of the variation in coal prices was explained by the model.

Using multiple metrics gave a more balanced view of model performance. The detailed model comparison and final model selection are presented in the next section.

9. Model Comparison and Performance Evaluation

9.1 Model Performance Summary

After developing and testing both models, the final comparison was made on the 2025 test period. This period was kept unseen during model tuning and was used to check how well each model performed on future data.

XGBoost and SARIMAX were compared using the same evaluation metrics: MAE, RMSE, MAPE, and R² score. RMSE was used as the main selection metric because it gives higher penalty to larger forecasting errors, which is important in commodity price forecasting.

Evaluation Stage	MAE	RMSE	MAPE (%)	R ² Score
Validation	1.41	2.25	1.05	0.92
Time-Series CV Mean	25.81	44.09	11.33	0.62
Test	1.27	1.63	1.21	0.93

TABLE 9.1: XGBOOST EVALUATION SUMMARY

Evaluation Stage	MAE	RMSE	MAPE (%)	R ² Score
Validation	1.32	2.21	0.98	0.92
Time-Series CV Mean	2.40	5.53	1.48	0.99
Test	1.94	2.25	1.84	0.86

TABLE 9.2: SARIMAX EVALUATION SUMMARY

9.2 Final Test Performance Comparison

The final model decision was based on the 2025 test results because this period was not used during training or validation. XGBoost performed better than SARIMAX across all main test metrics.

Model	MAE	RMSE	MAPE (%)	R ² Score
XGBoost Test	1.27	1.63	1.21	0.93
SARIMAX Test	1.94	2.25	1.84	0.86

TABLE 9.3: FINAL TEST PERFORMANCE COMPARISON

XGBoost had the lower MAE, lower RMSE, lower MAPE, and higher R^2 score. This showed that XGBoost produced more accurate predictions on the unseen 2025 test period.

9.3 RMSE Comparison

RMSE was treated as the main comparison metric because coal price forecasting should avoid large prediction errors. A larger error can have a direct impact on procurement planning, contract timing, and cost estimation.

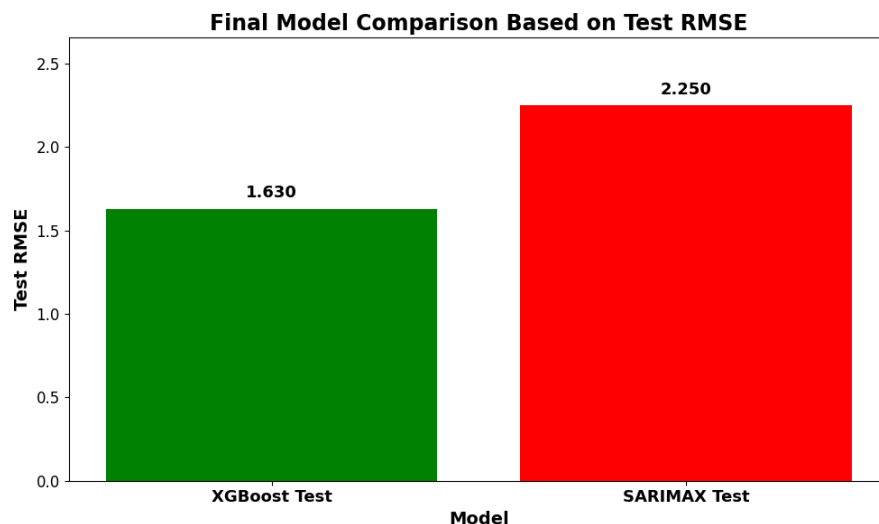


FIGURE 9.1: FINAL MODEL COMPARISON BASED ON TEST RMSE

XGBoost achieved a test RMSE of 1.63, compared with 2.25 for SARIMAX. This means XGBoost had fewer large errors on the test period and was more reliable for the final forecasting task.

9.4 MAE Comparison

MAE measures the average prediction error in coal price units. XGBoost achieved a test MAE of 1.27, while SARIMAX achieved a test MAE of 1.94.

This means that, on average, XGBoost predictions were closer to the actual 2025 coal prices. From a business point of view, lower MAE is useful because it gives a clearer view of expected pricing error in normal coal price units.

9.5 MAPE and R^2 Comparison

MAPE was used to understand the error in percentage terms. XGBoost achieved a test MAPE of 1.21%, while SARIMAX had a test MAPE of 1.84%. Both models performed reasonably well, but XGBoost had the lower percentage error.

R^2 was used to measure how well the model explained the variation in coal prices. XGBoost achieved an R^2 score of 0.93, compared with 0.86 for SARIMAX. This confirmed that XGBoost explained the test-period coal price movement better than SARIMAX.

9.6 Forecast vs Actual Test Results

The actual vs predicted comparison was reviewed on the 2025 test period. This helped check whether the models were producing realistic predictions, not only good metric values.

Date	Actual Coal Price	XGBoost Prediction	SARIMAX Prediction
2025-01-02	124.60	125.98	127.02
2025-01-03	123.50	124.20	126.12
2025-01-06	119.75	124.20	125.33
2025-01-07	118.30	120.19	121.40
2025-01-08	117.45	119.76	120.14

TABLE 9.4: SAMPLE FORECAST VS ACTUAL TEST RESULTS

The sample comparison shows that both models followed the general direction of coal prices, but XGBoost stayed closer to the actual values in the test period. This was also reflected in the lower MAE, RMSE, and MAPE values.

9.7 Strengths and Weaknesses of Each Model

Model	Strengths	Weaknesses
XGBoost	Captured non-linear relationships between coal prices and external market drivers; achieved the best test performance; handled calendar and lag features well	Depends on engineered features and is less directly interpretable than a traditional statistical model
SARIMAX	Provided a useful statistical time-series benchmark, used external variables as exogenous inputs; easier to explain as a classical forecasting method	Produced higher test error than XGBoost and was less flexible in capturing non-linear market behaviour

TABLE 9.5: MODEL STRENGTHS AND WEAKNESSES

Based on the final test comparison, XGBoost was selected as the final forecasting model. It gave the lowest RMSE, lowest MAE, lowest MAPE, and highest R^2 score on the 2025 test period. The selected XGBoost model was then used to generate the Newcastle coal price forecast for January to June 2026.

10. Final Model Selection and Justification

10.1 Best Performing Model

XGBoost was selected as the final forecasting model for Newcastle coal price prediction. The final decision was based on the 2025 test-period performance, where XGBoost performed better than SARIMAX across the main evaluation metrics.

The model produced lower MAE, RMSE, and MAPE, and also achieved a higher R^2 score compared with SARIMAX. Since the 2025 test period was kept unseen during training and validation, this result provided a fair basis for final model selection.

10.2 Why the Final Model Was Selected

XGBoost was selected because it handled the structure of this problem better than the statistical benchmark model. Coal prices are influenced by several market drivers at the same time, including energy prices, LNG, natural gas, currency movement, freight conditions, carbon prices, industrial commodity demand, and recent coal price behaviour.

The relationship between these variables and coal prices is not always linear. XGBoost was better suited for this type of feature-driven forecasting problem because it can capture non-linear patterns and interactions between variables. SARIMAX was useful as a traditional time-series benchmark, but its final test error was higher than XGBoost.

I 0.3 Model Assumptions

The final model assumes that historical relationships between Newcastle coal prices and the selected market variables provide useful signals for the January to June 2026 forecast period. It also assumes that recent coal price behaviour, represented through lag features, is relevant for short-term forecasting.

The model uses available trading-day records, not full calendar-day records. This is appropriate for market data because commodity and financial instruments do not trade on weekends and exchange holidays. Missing values caused by different market calendars were handled during data preparation before model development.

The model should be interpreted as a forecasting model, not as a causal economic model. The variables used in the model help predict coal price movement, but the model does not claim that any single variable directly causes coal prices to move.

I 0.4 Model Interpretability

XGBoost is less straightforward than a traditional statistical model, but it remains explainable at a practical level because the input features were clearly defined and selected based on market logic. The model used external market prices, calendar variables, and coal lag features, all of which have a clear business meaning.

The interpretation of the model was supported through EDA, correlation analysis, model performance comparison, and actual vs predicted review. This helped confirm that the final model was not treated as a black-box output only; it was evaluated against both statistical performance and market relevance.

I 0.5 Business Suitability of the Final Model

XGBoost was suitable for the business objective because it produced accurate daily forecasts and used market drivers that are relevant to coal procurement decisions. The model output can be summarized into monthly averages, which makes it easier to support procurement planning, contract timing, inventory decisions, and price risk monitoring.

For business users, the key advantage of the model is that it does not rely only on historical coal prices. It also considers broader market indicators such as energy prices, LNG, freight conditions, currency movement, and carbon pricing. This makes the forecast more useful for decision-making in a commodity market environment.

I 0.6 Trade-Offs Compared with Alternative Model

SARIMAX provided a useful statistical benchmark because it is a traditional time-series model and can include external variables. It is easier to explain from a classical forecasting perspective and is useful for comparison.

However, SARIMAX was less flexible in capturing non-linear relationships between coal prices and external market variables. XGBoost performed better on the unseen 2025 test period and was therefore selected as the final model.

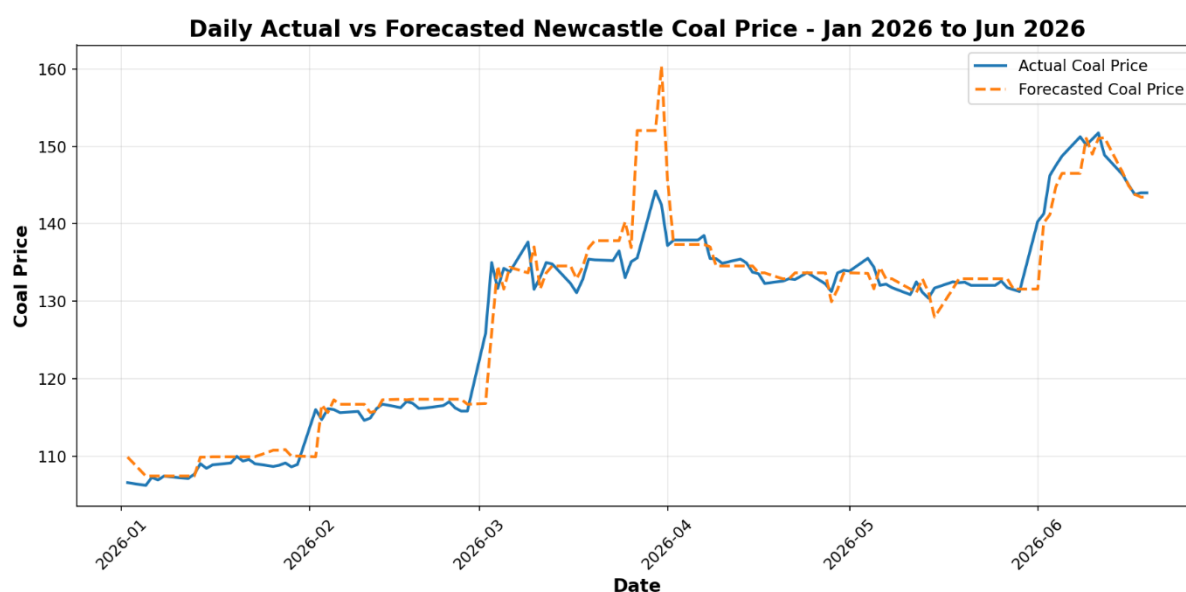
The trade-off is that XGBoost depends more on feature engineering and requires careful validation to avoid overfitting. This was addressed by using a chronological train-validation-test split, TimeSeriesSplit, and final evaluation on the unseen 2025 test period.

11. Coal Price Forecast Results: January 2026 - June 2026

11.1 Daily Forecast Output

The selected XGBoost model was used to generate daily Newcastle coal price forecasts for the January to June 2026 forecast-period records. The available forecast data covered 120 trading-day observations from 2 January 2026 to 19 June 2026. These records represent available market trading days, not full calendar days, because commodity and financial markets do not trade on weekends and exchange holidays.

The daily forecast was compared with the actual Newcastle coal price for the same available dates. This comparison helped check whether the final model was able to follow the real price movement during the forecast period.



Date	Actual Coal Price	Forecasted Coal Price	Selected Model
2026-01-02	106.55	109.86	XGBoost Test
2026-01-05	106.20	107.41	XGBoost Test
2026-01-06	107.25	107.41	XGBoost Test
2026-01-07	106.90	107.41	XGBoost Test
2026-01-08	107.40	107.41	XGBoost Test

TABLE 11.1: SAMPLE DAILY FORECAST OUTPUT

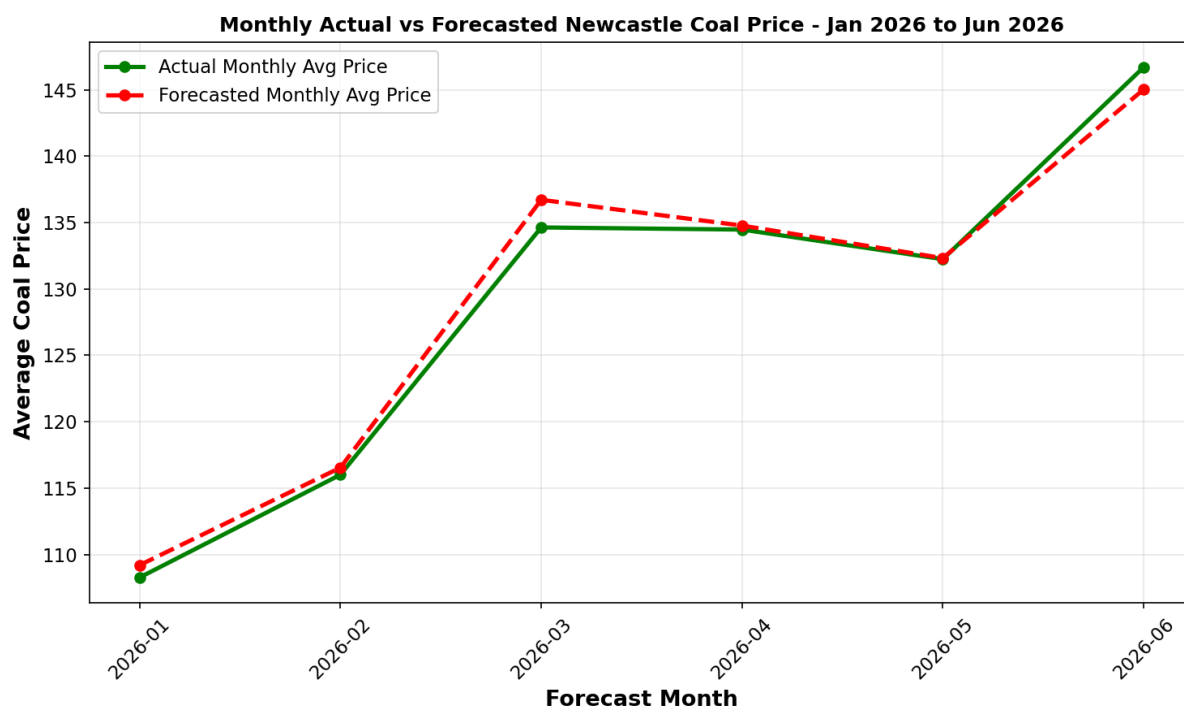
The daily forecast line followed the broad movement of the actual coal price series. The model captured the lower price range in January, the upward movement during February and March, the more stable movement during April and May, and the stronger price level seen in June. There were some short-term differences, especially around the late-March spike, but the overall direction was captured well.

11.2 Monthly Average Forecast Output

The daily forecast values were summarized into monthly averages because monthly-level interpretation is more useful for procurement planning and management reporting. The monthly table includes both actual and forecasted monthly average coal prices for the available 2026 records.

Forecast Month	Actual Monthly Average	Forecasted Monthly Average
2026-01	108.27	109.20
2026-02	116.01	116.54
2026-03	134.63	136.71
2026-04	134.47	134.77
2026-05	132.24	132.32
2026-06	146.69	145.02

TABLE 11.2: MONTHLY ACTUAL VS FORECASTED NEWCASTLE COAL PRICE



The monthly chart shows that the forecasted values were close to the actual monthly averages across the forecast period. The model slightly overestimated prices from January to April, was almost aligned in May, and slightly underestimated June. Overall, the monthly forecast captured the upward direction from January to June.

11.3 Forecast Trend Interpretation

The monthly forecast shows an overall upward movement from January to June 2026. The forecasted monthly average increased from 109.20 in January to 145.02 in June. The strongest increase appeared between February and March, where the forecast moved from 116.54 to 136.71.

April and May were relatively stable compared with March, while June showed the highest forecasted monthly average in the forecast period. This indicates that the model expected stronger coal prices toward the end of the forecast window.

11.4 Forecast Volatility Review

The daily forecast line was compared with the actual Newcastle coal price movement for the available 2026 trading records. The forecast followed the broad price direction and remained close to the actual values across most of the period.

The forecast did not attempt to capture every small day-to-day movement exactly. This is expected in commodity forecasting because daily market prices can move due to short-term news, trading activity, and temporary market shocks. For business use, the monthly forecast view is more stable and more useful for procurement planning than relying only on individual daily values.

11.5 Actual vs Predicted Comparison

The actual vs forecasted comparison showed that the selected XGBoost model performed well on the January to June 2026 available records. The forecasted monthly averages were close to the actual monthly averages in most months.

The model slightly overestimated January, February, March, and April, stayed almost equal in May, and slightly underestimated June. The difference was still reasonable for business-level planning, especially because the monthly forecast captured the overall upward direction correctly.

Forecast Month	Actual Monthly Average	Forecasted Monthly Average	Difference
2026-01	108.27	109.20	+0.93
2026-02	116.01	116.54	+0.53
2026-03	134.63	136.71	+2.08
2026-04	134.47	134.77	+0.30
2026-05	132.24	132.32	+0.08
2026-06	146.69	145.02	-1.67

TABLE 11.3: MONTHLY FORECAST DIFFERENCE

11.6 Key Forecast Takeaways

- The selected XGBoost model generated daily forecasts for 120 available trading-day records from January to June 2026.
- The daily forecast followed the broad direction of actual Newcastle coal price movement.
- The monthly forecast showed an upward price trend from January to June 2026.
- Forecasted monthly average coal price increased from 109.20 in January to 145.02 in June.
- March showed the largest monthly forecast difference, while May had the closest actual vs forecasted monthly average.
- June had the highest actual and forecasted price level in the available forecast-period records.
- The forecast output is suitable for business interpretation at the monthly level and is used in the next section for price optimization and procurement recommendations.

12. Price Optimization Framework

12.1 Purpose of Price Optimization

The purpose of the price optimization framework is to convert the coal price forecast into practical business decisions. Forecasting alone shows the expected price movement, but the business value comes from using that forecast for procurement timing, contract planning, inventory decisions, and risk management.

The January to June 2026 forecast showed an upward price movement, with forecasted monthly average prices increasing from 109.20 in January to 145.02 in June. This indicates that procurement decisions should not be delayed if the business has confirmed coal demand during the forecast period.

12.2 Procurement Planning Strategy

The forecast suggests that lower forecasted price months should be used for early or phased procurement. January and February showed relatively lower forecasted monthly prices compared with March and June.

From a procurement point of view, this creates an opportunity to secure part of the required volume earlier instead of waiting until the higher-price months.

Month	Forecasted Monthly Average	Price View	Recommended Procurement Action
January 2026	109.20	Lower price zone	Secure base requirement early
February 2026	116.54	Moderate price zone	Continue phased procurement
March 2026	136.71	Higher price zone	Avoid large spot exposure
April 2026	134.77	Higher price zone	Use only need-based spot buying
May 2026	132.32	Slightly lower than March/April	Review opportunity for limited top-up
June 2026	145.02	Highest forecast zone	Avoid unplanned purchases where possible

TABLE 12.1: FORECAST-BASED PROCUREMENT ACTION PLAN

12.3 Contract Timing Recommendation

Based on the forecasted upward trend, the preferred contract strategy is to lock a larger portion of expected demand during the lower forecasted price months. January and February appear more suitable for early contract locking or phased contract execution. Waiting until March or June could increase cost exposure if the forecasted upward movement materializes.

For buyers with flexible procurement schedules, the recommended approach is to split procurement into planned tranches rather than buying the full volume at one point. This reduces the risk of entering the market during a high-price period.

12.4 Inventory Management Recommendation

The forecast supports a higher inventory build-up during lower forecasted price periods, especially if storage capacity and working capital allow it. Building inventory in January and February can help reduce exposure to higher expected prices later in the forecast window.

However, inventory decisions should also consider storage cost, quality degradation risk, cash flow impact, and actual consumption plans. The forecast should therefore be used as a planning signal, not as the only decision factor.

12.5 Hedging Strategy Recommendation

For companies exposed to coal price volatility, the forecast can support hedging discussions. Since the forecast shows stronger prices toward the later part of the period, the business may consider partial hedging or fixed-price contracts for expected future demand.

The recommended hedging approach is not to hedge the full exposure at once. A phased hedge strategy is more practical, where part of the exposure is covered early, and the remaining exposure is reviewed as new market data becomes available. This helps balance cost protection with flexibility.

12.6 Scenario-Based Price Bands

A simple price-band framework was created to translate the forecast into action. The bands are based on the forecasted monthly average prices for January to June 2026.

Forecast Price Band	Interpretation	Business Action
Below 115	Lower price zone	Prioritize buying or contract locking
115 to 135	Moderate price zone	Use phased procurement and monitor market movement
Above 135	Higher price zone	Limit spot exposure and review hedging options

TABLE 12.2: PRICE BAND AND DECISION FRAMEWORK

Using this framework, January falls into the lower price zone, February and May fall closer to the moderate zone, while March, April, and June require more caution because prices are forecasted at higher levels.

12.7 Risk-Based Decision Rules

The following decision rules can be used by procurement and risk teams:

- If forecasted prices are in the lower zone, secure base demand early through spot purchase or short-term contract locking.
- If forecasted prices are in the moderate zone, buy in phases and continue monitoring external drivers such as Brent oil, natural gas, JKM LNG, freight rates, and the US Dollar Index.
- If forecasted prices are in the higher zone, avoid large unplanned spot purchases and consider fixed-price contracts or hedging support.
- If forecasted prices rise month over month, prioritize procurement earlier rather than delaying purchases.
- If the market moves against the forecast, update the forecast using the latest data before making large procurement decisions.

Overall, the optimization framework recommends early and phased procurement during lower forecasted price months, limited spot exposure during higher forecasted months, and continuous monitoring of energy, freight, currency, and LNG indicators. This approach helps convert the forecast into a practical cost-control and risk-management strategy.

13. Business Recommendations

13.1 Short-Term Procurement Recommendation

Recommendation: Secure part of the confirmed coal requirement during the lower forecasted price months, mainly January and February 2026.

Business actions:

- Lock the base procurement requirement early instead of leaving the full demand open for later spot purchases.
- Avoid purchasing the entire six-month requirement at once, as this can increase storage cost, working capital pressure, and inventory risk.
- Use phased procurement so the business gets cost protection while still keeping flexibility for market changes.
- Treat January and February as better procurement windows compared with the higher forecasted price months later in the period.

13.2 Medium-Term Contracting Recommendation

From March onwards, the forecast indicates a higher price zone compared with January and February. For this reason, large unplanned spot purchases should be avoided during March to June unless there is a clear operational need.

A balanced contract structure is recommended:

Contract Component	Recommended Action	Business Reason
Fixed-price portion	Lock confirmed base demand	Protects the business if prices continue to rise
Index-linked portion	Use for flexible demand	Allows adjustment if market prices soften
Spot purchase portion	Keep limited and controlled	Reduces exposure during high-price months

This approach is more practical than taking a fully fixed or fully spot-based position. It gives the business cost protection for essential demand while keeping some flexibility for changing market conditions.

13.3 Risk Management Recommendation

Recommendation: Link procurement decisions with forecast price bands and market signals.

Business actions:

- Set internal price review triggers based on the forecasted monthly price bands.
- Reduce spot exposure when forecasted prices move into the high-price zone.
- Review fixed-price contracts or hedging options if March to June price pressure continues.
- Avoid making large procurement decisions based only on one forecast run.
- Monitor the key market indicators used in the model: JKM LNG, natural gas, Brent oil, Baltic Dry Index, US Dollar Index, AUD/USD, carbon emissions, and iron ore.

13.4 Inventory Management Recommendation

Recommendation: Build inventory only when the forecasted price advantage is higher than storage and working capital cost.

Business actions:

- Maintain minimum safety stock to avoid operational disruption.
- Consider additional stock build-up during lower forecasted price months if storage capacity is available.
- Avoid excessive inventory if demand is uncertain or carrying cost is high.
- Use the forecast as a planning signal, but validate it against actual consumption plans before committing large volumes.

13.5 Monitoring Recommendation

Coal prices can change quickly because of energy market movement, freight conditions, currency shifts, policy changes, and supply-demand shocks. The forecast should therefore be refreshed regularly instead of being treated as a one-time output.

Recommended monitoring approach:

- Review the forecast monthly during normal market conditions.
- Move to weekly review during high-volatility periods.
- Refresh the forecast before any major purchase, contract renewal, or hedging decision.
- Track LNG, natural gas, Brent oil, freight rates, USD movement, AUD/USD, and carbon prices as early warning indicators.

13.6 Final Business Interpretation

The final recommendation is to use the forecast as a decision-support tool for procurement timing and cost control. The forecast indicates that delaying procurement into the higher forecasted months may increase cost exposure. Therefore, the business should secure the base requirement early, use split contracts for medium-term exposure, limit unplanned spot purchases during high-price months, and update the forecast as new market data becomes available.

The recommended strategy is not aggressive buying or full hedging. The stronger approach is a balanced base-flex-spot structure: lock confirmed demand early, keep flexible demand under controlled contract terms, and reserve spot buying only for short-term operational needs.

14. Reproducibility and Technical Implementation

The project was developed in Python using Google Colab. Google Drive was mounted to access the raw CSV files and save the cleaned datasets, model-ready dataset, forecast outputs, and supporting files.

The main libraries used were pandas, numpy, matplotlib, seaborn, scikit-learn, xgboost, and statsmodels. The notebook follows a sequential workflow from data loading, cleaning, EDA, feature engineering, model development, model comparison, forecasting, and price optimization

The notebook was organized in a sequential workflow: project setup, data loading, data cleaning, EDA, feature engineering, model development, model comparison, final forecasting, and price optimization. This structure makes the analysis easy to follow and allows the results to be reproduced by running the notebook from top to bottom.

The key output files generated from the workflow were:

Output File	Purpose
cleaned_master_dataset.csv	Cleaned and merged dataset used for EDA and modelling preparation
model_ready_coal_forecasting_data.csv	Final feature-ready dataset used for model development
final_newcastle_coal_price_forecast_jan_jun_2026.csv	Final forecast output containing actual prices, forecasted prices, monthly averages, selected model, and input variables

14.5 Reproducibility Note

To reproduce the results, the raw CSV files should be placed in the expected source folder, and the notebook should be executed in order. The notebook performs the complete workflow from raw data loading to final forecast output generation.

15. Assumptions, Limitations, and Future Enhancements

15.1 Key Assumptions

The model uses Newcastle Coal Futures as the target benchmark for coal price forecasting. Trading-day records were used instead of full calendar-day records because commodity and financial markets do not trade on every calendar day. Missing values created by different trading calendars were handled using forward fill, and Volume was excluded because it was not consistently available across datasets.

15.2 Data Limitations

The data was collected from publicly available CSV downloads. This made the project cost-effective and suitable for the assignment, but it does not provide the same automation, licensing support, or governance as an institutional data feed. The available forecast-period comparison covered trading records from 2 January 2026 to 19 June 2026.

15.3 Model Limitations

The selected XGBoost model performed better than SARIMAX on the 2025 test period, but it should still be used as a decision-support tool rather than a single source of truth. The model can capture broad price direction and monthly movement, but sudden market shocks, policy changes, supply disruptions, or geopolitical events may still affect actual prices.

15.4 Future Enhancements

Future improvements could include automated data ingestion, additional coal market fundamentals, inventory data, import-export data, weather or power demand indicators, forecast confidence intervals, and model explainability outputs such as feature importance or SHAP analysis.

16. Declaration, AI Assistance Disclosure, & Confidentiality

16.1 Use of GenAI Tools

AI tools were used for guidance in structuring the report, improving clarity, and reviewing presentation quality. The data preparation, modelling workflow, interpretation of results, assumptions, and business recommendations were reviewed and validated by me before final submission.

16.2 Candidate Ownership Statement

I confirm that I have prepared this assignment as part of the Wipro Data Scientist hiring process. The datasets were collected, cleaned, analysed, and modelled for the purpose of this submission. The final interpretation and recommendations are based on my review of the results. This report is intended only for the hiring assessment and should be treated as confidential.

17. Appendix

Model	Evaluation Stage	MAE	RMSE	MAPE (%)	R² Score
XGBoost	Validation	1.41	2.25	1.05	0.92
XGBoost	Time-Series CV Mean	25.81	44.09	11.33	0.62
XGBoost	Test	1.27	1.63	1.21	0.93
SARIMAX	Validation	1.32	2.21	0.98	0.92
SARIMAX	Time-Series CV Mean	2.40	5.53	1.48	0.99
SARIMAX	Test	1.94	2.25	1.84	0.86

17.1 FULL MODEL METRICS TABLE